

An Audited Baseline and Proof Interfaces for Erdős Problem 892

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Abstract

We give an audited research baseline and two new partial advances for the questions known as Erdős Problem 892. We prove the exact counting and compactness interfaces, an explicit reciprocal-load sufficient condition for arbitrary source irregularity, and the harmonic obstruction for prescribed density scales. For the quasi-primitive case we prove a mixed-rank two-shadow theorem and compress every Hasse palette by one prime deletion plus one abstract bit. For the logarithmic-barrier density profile we prove a uniform semiprime lower bound which rules out unconditioned anchorwise saturation, and we give a period-free Bonferroni transfer lemma. Strict counterexamples delimit the failed mechanisms. These results have been independently audited, but the arithmetic label-erasure and maximal-anchor actual-union steps remain open; no solution of either characterization is claimed.

1 The problem and its historical convention

A set $A \subseteq \mathbb{N}$ is *primitive* if no two distinct members divide one another. Write $A(x) = |A \cap [1, x]|$.

Question 1.1 (Domination). For which strictly increasing integer sequences $b_1 < b_2 < \dots$ do there exist a primitive sequence $a_1 < a_2 < \dots$ and $C < \infty$ such that

$$a_n \leq Cb_n \quad (n \geq 1)?$$

Question 1.2 (Prescribed density scales). For which strictly increasing exponent sequences $n_1 < n_2 < \dots$ do there exist a primitive set A and $c > 0$ such that

$$A(2^{n_i}) \geq c2^{n_i} \quad (i \geq 1)?$$

The original conference paper states the special hypothesis $\gcd(b_i, b_j) \neq b_r$ only for incomparable b_i, b_j [4, pp. 41–42]. We therefore make the following definition.

Definition 1.3. A set $B \subseteq \mathbb{N}$ is *quasi-primitive* if

$$x, y, d \in B, \quad x \nmid y, \quad y \nmid x \implies \gcd(x, y) \neq d.$$

If the phrase “no non-trivial solution” were instead read as excluding every gcd identity except $i = j = r$, then $b_i \mid b_j$ for $i \neq j$ would already be impossible, so B itself would be primitive. That literal reading makes the special case trivial and is not the historical question.

2 Exact domination interfaces

Let $B(x) = |\{n : b_n \leq x\}|$.

Proposition 2.1 (Counting formulation). *For an increasing sequence $A = \{a_1 < a_2 < \dots\}$ and fixed $C \geq 1$, the following are equivalent:*

- (a) $a_n \leq Cb_n$ for every n ;
- (b) $A(Cx) \geq B(x)$ for every real $x \geq 1$;
- (c) $A(Cb_n) \geq n$ for every n .

Proof. If $B(x) = m$, then $b_m \leq x$, hence $a_m \leq Cb_m \leq Cx$ and $A(Cx) \geq m$. Thus (a) implies (b), while (b) implies (c) by taking $x = b_n$. Finally (c) says that the n th element of A is at most Cb_n , proving (a). $\square$

Proposition 2.2 (Finite-prefix compactness). *Fix an integer $C \geq 1$. There exists an infinite primitive sequence with $a_n \leq Cb_n$ for all n if and only if, for every N , there is a primitive tuple*

$$c_1 < \dots < c_N, \quad c_i \leq Cb_i \quad (1 \leq i \leq N).$$

Proof. Necessity follows by truncation. Conversely, form a rooted tree whose level- N vertices are the feasible N -tuples, with edges given by extension. Every vertex has finitely many children because the next coordinate is at most Cb_{N+1} , and every level is nonempty by hypothesis. König's infinity lemma gives an infinite branch; its union is the required primitive sequence. $\square$

Proposition 2.2 is an exact necessary-and-sufficient condition but not yet an arithmetic characterization. For the quasi-primitive special case, the missing assertion is a uniform bound on the finite dilation constants, not merely the existence of a witness for each fixed prefix with a prefix-dependent constant.

3 A sufficient theorem for arbitrary irregularity

The following result appeared in the supplied manuscript and has now received two independent line-by-line audits. Its proof is included in full.

Theorem 3.1 (Dyadic reciprocal-load criterion). *Let $b_1 < b_2 < \dots$ be positive integers and suppose*

$$H := \sum_{n=1}^{\infty} \frac{1}{b_n} < \infty.$$

If K is an integer with $K \geq \max\{4, 4H\}$, then there is a primitive sequence $a_1 < a_2 < \dots$ satisfying

$$a_n \leq 2Kb_n \quad (n \geq 1).$$

Proof. For $j \geq 0$ put

$$D_j = \{n : 2^j \leq b_n < 2^{j+1}\}, \quad m_j = |D_j|, \quad I_j = (K2^j, 2K2^j] \cap \mathbb{N}.$$

Distinctness of the b_n gives $m_j \leq 2^j$, while I_j has exactly $K2^j$ members. We recursively choose m_j targets from I_j .

Suppose all earlier target blocks have been selected and let $E_{<j}$ be their union. Since a target chosen from I_i exceeds $K2^i$,

$$R_{<j} := \sum_{a \in E_{<j}} \frac{1}{a} < \frac{1}{K} \sum_{i < j} \frac{m_i}{2^i}.$$

For $n \in D_i$ we have $b_n < 2^{i+1}$, and consequently

$$\frac{m_i}{2^i} < 2 \sum_{n \in D_i} \frac{1}{b_n}.$$

Thus $R_{<j} < 2H/K \leq 1/2$. We also have

$$|E_{<j}| = \sum_{i < j} m_i \leq \sum_{i < j} 2^i = 2^j - 1.$$

Delete from I_j every multiple of an earlier target. A fixed $a \in E_{<j}$ has at most $K2^j/a + 1$ multiples in this interval, so the union bound deletes at most

$$K2^j R_{<j} + |E_{<j}| \leq \frac{K}{2} 2^j + 2^j - 1.$$

At least

$$K2^j - \frac{K}{2} 2^j - 2^j + 1 = \left(\frac{K}{2} - 1\right) 2^j + 1 \geq 2^j \geq m_j$$

candidates remain. Choose any m_j of them.

Each target block is primitive because two distinct integers in $(X, 2X]$ cannot divide one another. Every later block is strictly larger than every earlier block; size excludes divisibility from later to earlier, while the deletion rule excludes divisibility from earlier to later. The union is therefore primitive. Finally, the source bins and target blocks occur in the same order. Matching within each block increasingly, $n \in D_j$ implies

$$a_n \leq 2K2^j \leq 2Kb_n.$$

□

This criterion permits arbitrary burstiness. It is not necessary: a sequence of order $n \log n$ is dominated by the primes although its ordinary reciprocal sum diverges.

4 Published necessary conditions

Erdős proved that every primitive A satisfies

$$\sum_{\substack{a \in A \\ a > 1}} \frac{1}{a \log a} < \infty$$

[2]. Erdős–Sárközy–Szemerédi later proved that every infinite primitive A satisfies

$$\sum_{\substack{a \in A \\ a < x}} \frac{1}{a} = o\left(\frac{\log x}{\sqrt{\log \log x}}\right)$$

[3]. Applying these to a dominator and using $a_n \leq Cb_n$ gives the two classical necessary conditions displayed on the problem page:

$$\sum_n \frac{1}{b_n \log b_n} < \infty, \quad \sum_{b_n < x} \frac{1}{b_n} = o\left(\frac{\log x}{\sqrt{\log \log x}}\right).$$

The comparison is unaffected by finitely many terms or by replacing x with Cx .

Martin and Pomerance prove a substantial approximate converse for smoothly varying counting profiles [7]. It does not coordinate arbitrary abrupt bursts, which are central to both questions here.

5 A necessary condition for prescribed scales

Theorem 5.1. *If a primitive set A and a constant $c > 0$ satisfy $A(2^{n_i}) \geq c2^{n_i}$ for every i , then*

$$\sum_i \frac{1}{n_i} < \infty.$$

Proof. We may take $0 < c \leq 1$. Put $X_i = 2^{n_i}$ and $\delta = c/2$. For all sufficiently large i ,

$$|A \cap (\delta X_i, X_i]| \geq A(X_i) - \lfloor \delta X_i \rfloor \geq \frac{c}{3} X_i.$$

Every element of this slice contributes at least $(X_i \log X_i)^{-1}$ to the Erdős sum, so the slice contributes at least

$$\frac{c}{3 \log X_i} = \frac{c}{3(\log 2)n_i}.$$

Choose an integer r with $\delta 2^r > 1$. Since the n_i are strictly increasing integers, the slices with indices in any fixed residue class modulo r are pairwise disjoint. If $\sum_i 1/n_i$ diverged, one residue-class subseries would diverge and force the Erdős sum of A to diverge, a contradiction. $\square$

6 The supplied partial-results manuscript

The 169-page stage manuscript [1] explicitly states that it does not solve either characterization or the quasi-primitive special case. It proposes many strong partial theorems. The present project treats them as candidates rather than inherited facts until their hypotheses, analytic citations, and proofs are independently checked. The terminal interfaces extracted from its dependency ledger, sharpened by the audited results below, are:

1. for quasi-primitive inputs, an arithmetic erasure of the Hasse parent/type labels and cross-rank divisibility with uniformly bounded dilation, or a genuine divergent obstruction;
2. a necessary-and-sufficient cross-scale condition for arbitrary domination beyond scalar and consecutive-load restrictions;
3. a converse, or further obstruction, for exponent sequences satisfying harmonic, global, and uniform local crowding restrictions;
4. an actual-union estimate after maximal-anchor conditioning for middle-anchor low-multiplier events in a terminal short interval; the unconditioned supremum and mean anchorwise estimates are proved below to be too large.

The most concrete first-question finite target is the conjectural universal factor-two witness for every finite quasi-primitive tuple. Finite verification cannot by itself prove the infinite assertion, but a uniform theorem would combine immediately with Proposition 2.2. Conversely, a single finite tuple requiring a constant greater than two would refute only the factor-two strengthening, not the original bounded-constant question.

7 Mixed-rank shadows for quasi-primitive Hasse palettes

For $m \geq 1$ let $\Omega(m) = \sum_p v_p(m)$, let $\text{supp}(m) = \{p : p \mid m\}$, and, for $E \subseteq \{2, 3, \dots\}$, put

$$\partial E = \{e/p : e \in E, p \text{ prime}, p \mid e\}.$$

We use the following published form of Katona's intersecting-shadow theorem: if $\mathcal{F}$ is a finite intersecting family of s -subsets, then

$$|\partial_{\text{set}} \mathcal{F}| \geq |\mathcal{F}|. \quad (\text{KS})$$

See [6]; the more general statement in [5, Theorem 1.3], with $t = 1$ and $\ell = s - 1$, also gives (KS). The case $s = 1$ is immediate. Notice that (KS) applies to every subfamily of an intersecting family.

Theorem 7.1 (Mixed-rank two-shadow map). *Let $E \subseteq \{2, 3, \dots\}$ be finite or countable and satisfy $\gcd(e, f) > 1$ for all distinct $e, f \in E$. There are maps*

$$\psi : E \longrightarrow \mathbb{N}, \quad \tau : E \longrightarrow \{0, 1\},$$

such that $e/\psi(e)$ is prime and

$$e \longmapsto (\psi(e), \tau(e)) \quad (\text{MS})$$

is injective. Equivalently, one may choose an immediate multiplicative shadow of each member so that every integer is chosen at most twice. Hence $|G| \leq 2|\partial G|$ for every finite $G \subseteq E$.

Proof. First suppose that E is finite and all its members have the same value $\Omega(e) = k$. Fix a support size s and set

$$\mathcal{A}_s = \{\text{supp}(e) : e \in E, |\text{supp}(e)| = s\}.$$

This set family is intersecting. By (KS), every $\mathcal{G} \subseteq \mathcal{A}_s$ satisfies $|\partial_{\text{set}} \mathcal{G}| \geq |\mathcal{G}|$; Hall's theorem therefore gives an injection

$$T_s : \mathcal{A}_s \longrightarrow \partial_{\text{set}} \mathcal{A}_s, \quad T_s(S) \subset S, \quad |S \setminus T_s(S)| = 1.$$

Let $q_s(S)$ be the unique prime in $S \setminus T_s(S)$ and define $\psi_s(e) = e/q_s(S)$ when $\text{supp}(e) = S$.

The map ψ_s is injective on this support-size slice. Indeed, if the deleted prime occurs in e with exponent at least two, the support of the image is S ; if it occurs with exponent one, the support of the image is $T_s(S)$. Two images of the first kind determine the same S , and two of the second kind determine the same S by injectivity of T_s ; in either case the deleted prime and then e are determined. Images of the two kinds cannot coincide because their support sizes are s and $s - 1$.

Now fix an output d with support size r . It can have a preimage only from the support-size r slice, when a repeated prime was deleted, or from the support-size $r + 1$ slice, when a prime disappeared. Each slice supplies at most one preimage, so the combined map has fibres of size at most two. For a finite mixed-rank family, apply this construction separately to $E_k = \{e \in E : \Omega(e) = k\}$. Its images all have rank $k - 1$, so images coming from different k do not meet. Colour the at most two members of each fibre by 0 and 1 to obtain (MS).

Finally let E be countable. Make a bipartite graph with left side E , right side $\mathbb{N} \times \{0, 1\}$, and an edge $e \sim (d, t)$ exactly when e/d is prime. Every left degree is finite. The finite case verifies Hall's condition for every finite left subset. Matching each finite initial segment and applying König's infinity lemma to the resulting finitely branching tree gives a matching of all of E , which is (MS). Restriction to a finite G gives $|G| \leq 2|\partial G|$. $\square$

This theorem applies canonically to the local Hasse palettes of a quasi-primitive set. In the divisibility poset induced by B , let $L_B(b)$ denote the lower covers of b . Every nonminimal b has such a cover because b has only finitely many divisors. Choose $\sigma(b) \in L_B(b)$ for every nonminimal b , let R be the minimal members of B , and put

$$E_d = \{b/d : \sigma(b) = d\}.$$

Corollary 7.2 (Hasse palette compression). *If B is finite or countable and quasi-primitive, every E_d is primitive and pairwise non-coprime. Moreover, there are maps $s : B \setminus R \rightarrow \mathbb{N}$ and $t : B \setminus R \rightarrow \{0, 1\}$ such that*

$$b \mapsto (\sigma(b), s(b), t(b)) \quad \text{is injective,} \quad \sigma(b)s(b) = \frac{b}{p_b} \leq \frac{b}{2} \quad (\text{HC})$$

for a prime $p_b \mid b/\sigma(b)$.

Proof. If distinct $e_1, e_2 \in E_d$ satisfied $e_1 \mid e_2$, then $d < de_1 < de_2$ and $d \mid de_1 \mid de_2$, contrary to d being a lower cover of de_2 . Thus E_d is primitive. If $\gcd(e_1, e_2) = 1$, the preceding conclusion makes de_1, de_2 incomparable, while $\gcd(de_1, de_2) = d \in B$, contradicting quasi-primitivity. Apply Theorem 7.1 to each E_d , and for $b = de$ set $s(b) = \psi_d(e)$ and $t(b) = \tau_d(e)$. The parent label separates the palettes, (MS) gives injectivity within each palette, and deleting one prime gives (HC). $\square$

The binary type in this statement is not a proof artefact.

Proposition 7.3 (Strict one-type obstruction). *For distinct primes p, q, r , the family*

$$E^* = \{qr^2, q^2r, pr^2, pqr, pq^2, p^2r, p^2q\}$$

is primitive and pairwise non-coprime, but it has no injective map $\varphi : E^ \rightarrow \partial E^*$ with $e/\varphi(e)$ prime.*

Proof. All seven members have $\Omega = 3$, so none divides another. Their prime supports are the three two-element subsets of $\{p, q, r\}$, together with $\{p, q, r\}$, and hence meet pairwise. Directly,

$$\partial E^* = \{p^2, q^2, r^2, pq, pr, qr\}.$$

Thus an immediate-shadow injection would map seven objects into six. $\square$

Proposition 7.3 excludes every universal one-type immediate-shadow scheme; it does not show that the coefficient 2 in the finite inequality is the best possible real constant. More importantly, Corollary 7.2 is still only an *abstract labelled* compression. Shadow values from different ranks can be comparable (for example, the valid choices $6 \mapsto 2$ and $20 \mapsto 4$), and labels attached to different parents can collide. The unproved step is a bounded-cost arithmetic erasure of the parent and type labels that leaves a primitive ordered sequence. Consequently these results do not settle the quasi-primitive case of Problem 892.

8 Moving-endpoint avoidance at the logarithmic barrier

Consider the test profile

$$n_j = \lceil j + j(\log(j+1))^2 \rceil, \quad X_j = 2^{n_j}.$$

For $i < k$, an odd $d \in (X_i/2, X_i]$, and $1 \leq h \leq i$, put

$$Y = Y_{i,k,d} = X_k/d, \quad D_{i,h}(d) = (X_{i+h}/(2d), X_{i+h}/d] \cap \mathbb{N}_{\text{odd}},$$

and define the unconditioned finite-block avoidance set

$$E_{i,k}^{\square}(d) = \left\{ r \in (Y/2, Y] \cap \mathbb{N}_{\text{odd}} : a \nmid r \text{ for all } a \in \bigcup_{h \leq i} D_{i,h}(d) \right\}. \quad (\text{A})$$

For a middle anchor $i < k/2$, this is the quotient set appearing in the maximal-old-divisor reduction of the stage manuscript, but (A) omits both maximal-anchor survival and terminal-pool rank conditions.

The next result uses precisely these standard analytic inputs, uniformly as their arguments tend to infinity:

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + O(1), \quad \sum_{x < p \leq 2x} \frac{1}{p} \ll \frac{1}{\log(2x)}, \quad (\text{AI1})$$

$$\sum_{\substack{x < u \leq 2x \\ \Omega(u)=2}} \frac{1}{u} \ll \frac{\log \log(3x)}{\log(2x)}, \quad \pi(x) - \pi(x/2) \gg \frac{x}{\log x}. \quad (\text{AI2})$$

Here the first formula is Mertens' theorem. The second is the standard dyadic prime-reciprocal estimate (for example, a consequence of the prime number theorem), the first estimate in (AI2) is the standard dyadic semiprime upper bound, and the last is the prime number theorem in dyadic intervals. For completeness, the semiprime bound follows by writing $u = pq$ with $p \leq q$, applying the usual PNT upper bound to q for each $p \leq \sqrt{2x}$, and then using Mertens' estimate. No prime theorem in an interval shorter than a fixed positive proportion of its endpoint is used.

Proposition 8.1 (Moving-endpoint semiprime floor, modulo (AI1)–(AI2)). *Assume the named standard estimates (AI1)–(AI2). Fix $0 < a < b < 1/2$. There is $c_{a,b} > 0$ such that, for all sufficiently large k , all integers i with $ak \leq i \leq bk$, and every odd $d \in (X_i/2, X_i]$,*

$$|E_{i,k}^{\square}(d)| \geq c_{a,b} \frac{Y_{i,k,d}}{i}. \quad (\text{SF})$$

Proof. Write $Z = X_{2i}/d$ and $\mathcal{D} = \bigcup_{h \leq i} D_{i,h}(d)$. If $\Theta_h = X_{i+h}/(2d)$, then $D_{i,h}(d) = (\Theta_h, 2\Theta_h] \cap \mathbb{N}_{\text{odd}}$ and $\mathcal{D} \subset (1, Z]$. The mean-value theorem applied to $x + x(\log(x+1))^2$, with the ceiling errors absorbed, gives uniformly for $1 \leq h \leq i$

$$\log \Theta_h \asymp h(\log i)^2, \quad \log Z \asymp i(\log i)^2. \quad (1)$$

Regular variation, uniformly for $ak \leq i \leq bk$, gives

$$\frac{\log Y}{\log Z} = \frac{n_k - n_i + O(1)}{n_{2i} - n_i + O(1)} = \frac{k - i}{i} (1 + o_{a,b}(1)).$$

Since $(k-i)/i \geq (1-b)/b > 1$, there is a constant $\delta_{a,b} > 0$ such that, for all sufficiently large k ,

$$\frac{\log Y}{\log Z} \geq 1 + \delta_{a,b}. \quad (2)$$

Choose $0 < \eta < \min\{1, \delta_{a,b}/2\}$ and put $P = Z^{\eta/2}$. Let $\mathcal{V}$ be the products $u = pq$ of two distinct odd primes $p, q \leq P$ for which none of p, q, pq belongs to $\mathcal{D}$. By (AI1),

$$\sum_{\substack{p \leq P \\ p \text{ odd prime}}} \frac{1}{p} \asymp \log i. \quad (3)$$

On the other hand, (1), (AI1), and (AI2) imply

$$\sum_{h \leq i} \sum_{\Theta_h < p \leq 2\Theta_h} \frac{1}{p} \ll \frac{1}{\log i}, \quad (4)$$

and

$$\sum_{h \leq i} \sum_{\substack{\Theta_h < u \leq 2\Theta_h \\ \Omega(u)=2}} \frac{1}{u} \ll \frac{1}{(\log i)^2} \sum_{h \leq i} \frac{\log(3h(\log i)^2)}{h} \ll 1. \quad (5)$$

The reciprocal weight of all products of two distinct odd primes at most P is $\asymp (\log i)^2$. By (4), products for which one prime lies in $\mathcal{D}$ have total weight $O(1)$; by (5), the same is true for products which themselves lie in $\mathcal{D}$. Therefore

$$\sum_{u \in \mathcal{V}} \frac{1}{u} \gg (\log i)^2. \quad (6)$$

For $u \in \mathcal{V}$ we have $u \leq Z^\eta$, so (2) gives $x_u := Y/u \geq Z^{1+\delta_{a,b}/2}$. In particular $x_u/2 > Z$ for large k . The dyadic prime number theorem in (AI2), together with $\log x_u \asymp_{a,b} \log Z$, supplies

$$\#\{s : Y/(2u) < s \leq Y/u, \ s \text{ prime}\} \gg_{a,b} \frac{Y}{u \log Z}. \quad (7)$$

Every such s exceeds Z . Hence in $r = us$ the factor u is exactly the part supported on primes at most Z , so representations belonging to different u are distinct. Any divisor of r not exceeding Z divides $u = pq$; its possibilities are $1, p, q, pq$, none of which lies in $\mathcal{D}$. Thus every r counted in (7) belongs to (A). Summing (7) over u , and using (1) and (6), proves

$$|E_{i,k}^\square(d)| \gg_{a,b} \frac{Y}{\log Z} (\log i)^2 \gg_{a,b} \frac{Y}{i}.$$

□

It follows that the avoidance density for each middle anchor is at least $c_{a,b}/i$; consequently its sum over $ak \leq i \leq bk$ stays bounded away from zero. Thus neither a supremum-over- d anchorwise sum nor its unconditioned mean-over- d analogue can yield $o(1)$. This is a strict obstruction to that proof mechanism, not evidence that the profile is inadmissible: a terminal integer may be counted by many anchors, and the maximal-anchor and rank conditions omitted from (A) can change the actual union.

We finally record a transfer lemma that does not require a full period of the least common multiple. For $Y \geq 3$ let $I_Y = (Y/2, Y] \cap \mathbb{N}_{\text{odd}}$ and $N_Y = |I_Y|$. If F is a finite set of odd integers greater than one and $s \geq 1$, define

$$\mathcal{B}_{2s}(F) = 1 + \sum_{\substack{\emptyset \neq S \subseteq F \\ |S| \leq 2s}} (-1)^{|S|} \frac{1}{\text{lcm}(S)}.$$

Lemma 8.2 (Short-interval Bonferroni transfer). *Uniformly in Y, F, s ,*

$$\frac{1}{N_Y} |\{r \in I_Y : f \nmid r \text{ for every } f \in F\}| \leq \mathcal{B}_{2s}(F) + O\left(\frac{1}{Y} \sum_{j=0}^{2s} \binom{|F|}{j}\right). \quad (\text{BF})$$

If $2s \geq |F|$, inclusion-exclusion with the exact interval counts is an identity.

Proof. For every odd $q \geq 1$, direct counting gives, uniformly also for $q > Y$,

$$\frac{1}{N_Y} |\{r \in I_Y : q \mid r\}| = \frac{1}{q} + O(1/Y). \quad (8)$$

Apply the even Bonferroni upper bound to the complement of the union of the events $\{f \mid r\}$, $f \in F$. The intersection indexed by $S \subseteq F$ is $\{\text{lcm}(S) \mid r\}$. Substitution of (8) in the retained terms yields (BF), since their number is $\sum_{j \leq 2s} \binom{|F|}{j}$. Keeping every term gives the stated identity. $\square$

For a finite odd generator set F , write $E(F) = \{r \in \mathbb{N}_{\text{odd}} : f \nmid r \text{ for every } f \in F\}$.

Remark 8.3 (Conditional actual-union proof interface). The stage manuscript proposes a finite-block reduction with adjacent- and terminal-strip errors $o(X_k)$. That reduction has not been admitted as a theorem here. To state its remaining target without a free index, let $\mathcal{R}_k$ denote the finite set of pairs (i, d) retained by its maximal-anchor bookkeeping, where $i < k/2$ and $d \in (X_i/2, X_i]$ is odd. For each $(i, d) \in \mathcal{R}_k$, choose a finite

$$F_{i,k,d} \subseteq \bigcup_{1 \leq h \leq i} D_{i,h}(d).$$

The missing sufficient estimate inside that proposed reduction is

$$\lim_{J \rightarrow \infty} \sup_{k > J} \frac{1}{X_k} \left| \bigcup_{\substack{(i,d) \in \mathcal{R}_k \\ i \geq J}} d(I_{Y_{i,k,d}} \cap E(F_{i,k,d})) \right| = 0. \quad (\text{AU})$$

An actual quotient avoids the full generator union and hence every selected subset $F_{i,k,d}$, so (AU), together with the two asserted $o(X_k)$ error estimates, is exactly the terminal estimate required by that conditional argument. This paragraph records a proof interface; it neither proves (AU) nor promotes the unaudited finite-block reduction to a result of this paper.

Lemma 8.2 only transfers a finite certificate to the moving interval; it neither makes $\mathcal{B}_{2s}(F)$ small nor proves the actual-union estimate (AU). Proposition 8.1 rules out replacing that estimate by a sum of unconditioned anchorwise bounds. Thus neither admissibility nor inadmissibility of the logarithmic-barrier profile, and no solution of Problem 892, follows here.

9 Proof dependency and admission criteria

Any complete solution must supply a chain from one of the exact compactness interfaces to a verifiable arithmetic condition. A route is not discarded for lack of a standard tool. It is closed only by a counterexample, contradiction, or a theorem that its precise mechanism cannot imply the required node. Failed routes are retained as barrier theorems with their logical scope stated explicitly.

A final solution claim requires: (i) a complete paper proof; (ii) two independent adversarial reviews of every critical lemma; (iii) deterministic certificates for all finite computations; and (iv) a Lean theorem with no placeholders or project-defined axioms. None of these conditions is presently asserted to hold for the full problem.

10 Conclusion

The exact historical problem remains open in this repository. The audited advances show, on the first side, that every local quasi-primitive Hasse palette requires at most one binary bit under an immediate-shadow compression and, on the second side, that a broad anchorwise saturation strategy cannot resolve the logarithmic barrier. What remains is correspondingly precise: a globally primitive bounded-cost arithmetic encoding of the Hasse labels, and an actual-union estimate that exploits maximality or cross-anchor overlap. Subsequent branch work will be promoted only when it strengthens this ledger without silently assuming either open interface.

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